

Exact Sobolev nullity thresholds of fat Cantor sets

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Abstract

Hewett and Moiola asked whether a positive-measure set with empty interior can have Sobolev nullity threshold strictly between 0 and n/p , and asked for the threshold of their generalized Smith–Volterra–Cantor sets. We prove that, for every $n \geq 1$, $1 < p < \infty$, $0 < \alpha < 1/2$, and $0 < \beta < 1 - 2\alpha$,

$$s_{G_{\alpha,\beta}^{(n)}}(p) = \frac{1}{p} \left(1 + \frac{\log 2}{\log \alpha} \right).$$

The set is null at the threshold. The proof localizes the source paper’s Swiss-cheese non-nullity criterion, obtains the missing upper bound from one deleted gap at every level, and passes to Cartesian powers by directional Bessel-potential slicing.

1 The questions and statement

For $E \subset \mathbb{R}^n$, let $H_E^{s,p}$ be the distributions in $H^{s,p}(\mathbb{R}^n)$ supported by E . A set is (s,p) -null when $H_F^{s,p} = \{0\}$ for every closed $F \subset E$. For a set with empty interior, its nullity threshold $s_E(p)$ is characterized by nullity above the threshold and non-nullity below it [1].

The source questions are reproduced on the next page. In the notation of [1, Example 4.7], start with $l_0 = 1$ and put

$$l_{j+1} = \frac{1}{2}(l_j - \beta\alpha^j), \quad j \geq 0. \tag{1}$$

At stage $j+1$, remove from the middle of every one of the 2^j basic intervals a gap of length $\beta\alpha^j$. The remaining compact set is $G_{\alpha,\beta} \subset [0, 1]$, and $G_{\alpha,\beta}^{(n)}$ denotes its n -fold Cartesian power. Its measure is positive because

$$2^j l_j = 1 - \beta \frac{1 - (2\alpha)^j}{1 - 2\alpha} \longrightarrow L := 1 - \frac{\beta}{1 - 2\alpha} > 0. \tag{2}$$

Theorem 1 (Exact threshold). *Let $n \geq 1$, $1 < p < \infty$, $0 < \alpha < 1/2$, and $0 < \beta < 1 - 2\alpha$. Set*

$$\sigma_{\alpha,p} := \frac{1}{p} \left(1 + \frac{\log 2}{\log \alpha} \right). \tag{3}$$

Then

$$G_{\alpha,\beta}^{(n)} \text{ is } (s,p)\text{-null} \iff s \geq \sigma_{\alpha,p}.$$

In particular, $s_{G_{\alpha,\beta}^{(n)}}(p) = \sigma_{\alpha,p}$, and the set is null at its threshold.

Since $0 < \alpha < 1/2$, $0 < \sigma_{\alpha,p} < 1/p \leq n/p$. Theorem 1 therefore also answers the source’s existence question affirmatively in every dimension.

5 Conclusion

In Definition 2.1 and Proposition 2.4 we introduced the concepts of “ (s, p) -nullity” and the “nullity threshold” $s_E(p)$ of a subset $E \subset \mathbb{R}^n$, to describe the Sobolev regularity of the distributions supported by E . We now summarise our contributions to the questions **Q1–Q3** posed in §2.2. (An even more concise summary is given in tabular form in Table 2). We assume throughout this section that E is Borel with empty interior. Our first observation is that the case $m(E) = 0$ is significantly better understood than the case $m(E) > 0$.

Q1: Given $1 < p < \infty$ and $E \subset \mathbb{R}^n$ with empty interior, can we determine $s_E(p)$? If $m(E) = 0$ then $s_E(p)$ is immediately computed in terms of Hausdorff dimension as $s_E(p) = (\dim_H E - n)/p'$, see Theorem 2.12. If $m(E) > 0$ then we know $0 \leq s_E(p) \leq n/p$, but the only general result we know of that allows a more precise characterisation of $s_E(p)$ is Theorem 2.19, which appears useful only for proving lower bounds on $s_E(p)$. Using a result from [35] we described two types of set with maximum nullity threshold $s_E(p) = n/p$ for all $1 < p < \infty$, see Corollary 2.21 and Theorem 2.22. We also derived lower bounds on $s_E(p)$ for some “fat” and “super-fat” Cantor sets and Swiss cheese sets in §4.2.2 (the lower bound obtained for the fat Cantor set is represented by the dotted line in Figure 1). But we are unaware of any viable techniques for obtaining nontrivial upper bounds on $s_E(p)$ when $m(E) > 0$. In fact, we have no evidence whatsoever of the existence of sets with $s_E(p) \in (0, n/p)$.

Open question: Given $1 < p < \infty$, do there exist sets E for which $s_E(p) \in (0, n/p)$?

Open question: What is the nullity threshold of the fat Cantor sets of Example 4.7?

Q2: For which functions $f : (1, \infty) \rightarrow [-n, n]$ does there exist $E \subset \mathbb{R}^n$ such that $f(p) = s_E(p)$ for all $p \in (1, \infty)$? To describe the possible nullity-threshold functions, it is convenient to make the change of variable $p \mapsto r = 1/p$, and define, for $E \subset \mathbb{R}^n$ non-empty with empty interior,

$$S_E(r) := s_E(1/r) = \inf \{s \in \mathbb{R}, \text{ such that } E \text{ is } (s, 1/r)\text{-null}\}, \quad 0 < r < 1.$$

The function S_E satisfies the following conditions:

- $n(r-1) \leq S_E(r) \leq nr$ for $0 < r < 1$, so the graph of $S_E(r)$ in the rs -plane lies in the shaded parallelogram in Figure 1 (cf. Lemma 2.3(iv)–(vi)).
- S_E is non-decreasing and Lipschitz-continuous with $0 \leq S'_E(r) \leq n$ for a.e. $0 < r < 1$ (cf. Lemma 2.3(ii)).
- The graph of S_E in the rs -plane cannot cross the line $s = 0$. Precisely, if $S_E(r^*) \geq 0$ (or $S_E(r^*) < 0$) for some $r^* \in (0, 1)$ then $S_E(r) \geq 0$ (or $S_E(r) < 0$, respectively) for all $r \in (0, 1)$ (cf. Lemma 2.3(v)).
- If $m(E) = 0$ then $S_E(r) = (n - \dim_H E)(r - 1)$ (cf. Theorem 2.12). So if the graph of S_E lies in the lower half rs -plane in Figure 1, it is necessarily a straight line through $(1, 0)$ with slope equal to $n - \dim_H E$.

Hence the only functions $F : (0, 1) \rightarrow [-n, 0]$ which can be realised as $F(r) = S_E(r)$ for some $E \subset \mathbb{R}^n$ are straight lines $F(r) = \lambda(r - 1)$, $\lambda \in [0, n]$, which are realised by any set E with $m(E) = 0$ and $\dim_H E = n - \lambda$ (for example the Cantor sets of Theorem 4.5). By contrast, the only functions $F : (0, 1) \rightarrow [0, n]$ that we can provably realise as $F(r) = S_E(r)$ for some $E \subset \mathbb{R}^n$ are the two straight lines $F(r) = 0$ (already covered by the previous case), and the straight line $F(r) = nr$, with E given either by the compact set of Corollary 2.21 or the non-compact set of Theorem 2.22.

Figure 1: The two open questions in [1, p. 23].

2 Three tools

We use two results quoted and proved in the source paper. First, if $0 < s < 1/p$, the Bessel capacity of an interval of radius $r \leq 1$ satisfies

$$A_{s,p}r^{1-sp} \leq \text{cap}_{s,p}(B_r(x)) \leq B_{s,p}r^{1-sp} \quad (4)$$

with positive constants depending only on s, p ; see [1, Proposition 3.16]. Second, for a closed empty-interior $F \subset \mathbb{R}$, Theorem 2.19 of [1] gives

$$F \text{ is } (s, p)\text{-null} \iff \limsup_{\delta \downarrow 0} \frac{\text{cap}_{s,p}(B_\delta(x) \setminus F)}{\delta} > 0 \quad \text{for a.e. } x. \quad (5)$$

We also use the following sufficient condition, which is the one-dimensional case of [1, Theorem 4.6]. If

$$K = \overline{\Omega} \setminus \bigcup_i B_{r_i}(x_i)$$

is a compact Swiss-cheese set with empty interior, $B_r(x) \subset \Omega$, and $0 < s < 1/p$, then

$$\sum_i r_i^{1-sp} < \frac{A_{s,p}}{B_{s,p}} r^{1-sp} \implies K \text{ is not } (s, p)\text{-null}. \quad (6)$$

The final tool passes one-dimensional nullity to products.

Lemma 2 (Directional slicing). *Let $s \geq 0$, $1 < p < \infty$, and write $\mathbb{R}^n = \mathbb{R} \times \mathbb{R}^{n-1}$. Then*

$$H^{s,p}(\mathbb{R}^n) \hookrightarrow L^p(\mathbb{R}^{n-1}; H^{s,p}(\mathbb{R})). \quad (7)$$

Consequently, if $E \subset \mathbb{R}$ is closed and (s, p) -null, then E^n is (s, p) -null.

Proof. Let J_s be the Bessel lifting on $\mathbb{R}^n$ and $J_s^{(1)}$ the lifting in the first variable only. Applied to $J_s u$, the operator $J_s^{(1)} J_s^{-1}$ has multiplier

$$m_s(\xi, \eta) = \left(\frac{1 + |\xi|^2}{1 + |\xi|^2 + |\eta|^2} \right)^{s/2}.$$

On each coordinate orthant, direct differentiation shows that

$$\sup_{(\xi, \eta)} \left| \prod_{k: \gamma_k=1} \zeta_k \partial^\gamma m_s(\zeta) \right| < \infty, \quad \gamma_k \in \{0, 1\}.$$

The Marcinkiewicz multiplier theorem therefore gives $\|J_s^{(1)} u\|_p \lesssim \|J_s u\|_p$. Fubini is exactly (7).

If $u \in H^{s,p}(\mathbb{R}^n)$ is supported by E^n , then for almost every $y \in \mathbb{R}^{n-1}$ its slice $u(\cdot, y)$ belongs to $H^{s,p}(\mathbb{R})$ and vanishes almost everywhere off E . Its distributional support is therefore contained in E , so it is zero. Hence $u = 0$. $\square$

3 The exact one-dimensional threshold

Put $G = G_{\alpha, \beta}$ and $s_0 = \sigma_{\alpha, p}$. The critical arithmetic is

$$a_0 := 1 - ps_0 = -\frac{\log 2}{\log \alpha} \in (0, 1), \quad \alpha^{a_0} = \frac{1}{2}. \quad (8)$$

Nullity at the threshold

Fix $x \in G$. At level $j - 1$, choose a basic interval $I_{j-1}(x)$ of length l_{j-1} containing x . The central gap removed from it at level j has length

$$g_j = \beta \alpha^{j-1}.$$

Both x and this gap lie in $I_{j-1}(x)$, so $B_{2l_{j-1}}(x) \setminus G$ contains an interval of radius $g_j/2$. Since $l_{j-1} \leq 2^{-(j-1)}$, monotonicity and (4) give

$$\begin{aligned} \frac{\text{cap}_{s_0,p}(B_{2l_{j-1}}(x) \setminus G)}{2l_{j-1}} &\geq c \frac{g_j^{a_0}}{l_{j-1}} \\ &\geq c_{\alpha,\beta,p} (2\alpha^{a_0})^{j-1} = c_{\alpha,\beta,p} > 0. \end{aligned}$$

The radii $2l_{j-1}$ tend to zero. If $x \notin G$, then G^c is open; for all sufficiently small δ , the ball $B_\delta(x)$ is contained in G^c , and (4) makes the quotient in (5) tend to infinity. Thus (5) holds everywhere, and G is (s_0, p) -null. Standard Sobolev embeddings imply nullity for every $s \geq s_0$.

Non-nullity below the threshold

It is enough to consider $0 < s < s_0$; positive measure handles $s \leq 0$. Write

$$a := 1 - sp > 0, \quad q := 2\alpha^a.$$

By (8), $s < s_0$ is equivalent to $q < 1$.

Choose one level- J basic interval I_J and put $K_J = G \cap I_J$. Inside I_J , the global level- k gaps, $k > J$, have radius $g_k/2$ and multiplicity 2^{k-J-1} . Hence

$$S_J := \sum_{k>J} 2^{k-J-1} \left(\frac{g_k}{2}\right)^a \leq C_{\alpha,\beta,s,p} 2^{-J} q^J. \quad (9)$$

By (2), $l_J \geq L2^{-J}$. The interior of I_J contains a ball of radius $r_J = l_J/4$, and therefore

$$\frac{S_J}{r_J^a} \leq C_{\alpha,\beta,s,p} (2\alpha)^{aJ} \longrightarrow 0. \quad (10)$$

For all sufficiently large J , (10) is stronger than (6). Thus K_J , and hence G , is not (s, p) -null. This proves the one-dimensional case of Theorem 1.

4 Cartesian powers and consequences

At and above s_0 , Lemma 2 transfers the nullity of G to G^n . Conversely, let $s < s_0$ and choose t with $\max\{s, 0\} < t < s_0$. The one-dimensional result gives a nonzero $u \in H^{t,p}(\mathbb{R})$ supported by G . The tensor-product estimate recorded in [1, Proposition A.1] gives $u^{\otimes n} \in H^{s,p}(\mathbb{R}^n)$; it is nonzero and supported by G^n . For $s \leq 0$, the positive measure of G^n also gives non-nullity directly. This completes the proof of Theorem 1.

Corollary 3 (Intermediate thresholds in every dimension). *For every $n \geq 1$ and $1 < p < \infty$, there is a compact positive-measure empty-interior set $E \subset \mathbb{R}^n$ satisfying $0 < s_E(p) < n/p$. Indeed any $G_{\alpha,\beta}^{(n)}$ does so.*

Remark 4. *For the classical Smith–Volterra–Cantor choice $\alpha = \beta = 1/4$, Theorem 1 gives $s_{G_{1/4,1/4}^{(n)}}(p) = 1/(2p)$, and threshold nullity. The parameter β changes the measure and constants but not the exponent.*

5 Novelty and review note

A bounded search on 11 August 2026 covered the run registry, the exact paper title, the phrase “nullity threshold,” and the two quoted questions. It found the source and bibliographic copies, but no later primary paper stating Theorem 1. The proof uses the source’s own capacity tools in a localized way, so novelty confidence is moderate rather than definitive. Specialist review should focus first on Lemma 2 and on the critical use of (5).

References

- [1] D. P. Hewett and A. Moiola, *On the maximal Sobolev regularity of distributions supported by subsets of Euclidean space*, Analysis and Applications 15 (2017), 731–770; arXiv:1507.02698.